

EDA Summary Report

Project Title

Diamond Price Analysis using Exploratory Data Analysis

Objective

The objective of this project was to analyze the Diamonds dataset using Exploratory Data Analysis (EDA). I explored the dataset, performed statistical analysis, detected outliers, identified relationships between different variables, and created visualizations to understand the factors affecting diamond prices.

Dataset Description

The dataset contains information about diamonds, including their carat weight, cut, color, clarity, dimensions, and price. These features help in analyzing the factors that influence the price of a diamond.

Data Exploration

I explored the dataset using `head()`, `info()`, `shape`, `columns`, and `describe()`. This helped me understand the dataset structure, data types, and summary statistics.

Data Cleaning

I checked the dataset for missing values and duplicate records.

- No missing values were found.
- No duplicate records were found.

Therefore, no additional data cleaning was required.

Descriptive Statistics

I calculated the following statistical measures:

- Mean
- Median

- Mode
- Variance
- Standard Deviation

These measures helped me understand the distribution and spread of the numerical data.

Correlation Analysis

I calculated the correlation matrix to identify the relationship between different numerical features.

Observations

- Carat showed a very strong positive correlation (**0.929**) with price.
- The dimensions **x, y, and z** also had strong positive correlations with price.
- Depth had almost no correlation with price.
- The index column (Unnamed: 0) was not useful for analysis and was removed.

Outlier Detection

I detected outliers in the **price** column using the Interquartile Range (IQR) method. The outliers were treated using the capping method to reduce their impact on the analysis.

Data Analysis

I performed filtering and sorting operations to examine diamonds based on price and other important features.

Visualizations

The following visualizations were created:

- Bar Chart
- Line Chart
- Histogram
- Scatter Plot

These graphs helped me understand the distribution of prices and the relationship between different diamond features.

Conclusion

This project helped me improve my understanding of Exploratory Data Analysis and data preprocessing. I learned how to explore a dataset, calculate descriptive statistics, identify correlations, detect outliers, and create visualizations. Based on my analysis, **carat is the most important feature influencing diamond price**, while **depth has very little effect on price**.